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12T – Automated Fluid Dispenser Final Design Proposal

APSC 101

Team 141

Due: December 3rd, 2023

All work is ours except for cited materials.

Signed by: Ishaan, Nolan, Rafael, and Amarachukwu

1. Executive Summary

This report will uncover the specifics of how the group met the design goals, and how the design goals affect the experience of stakeholders, and users. The primary purpose of the automated fluid dispenser is to prevent overdoses, by automating a way to mix a solution of a powder in a saline solution in specific ratios. The precision of the ratios for the powder to fluid are necessary to reverse the effects of opioid drugs, so the dispenser will be expected to distribute these ratios accurately. The approach taken to tackle this problem is to create a turntable that will rotate five test tube slots, stopping when each test tube is under the dispensing head, until it has the amount of powder and fluid specified. Once this test tube is filled the turntable rotates and continues the same process until all remaining test tubes are filled.

Despite a simple approach, there are a lot of parts to the dispenser called sub-processes that require planning and modeling to make sure all parts are coordinated. These sub-processes include rotation of the turntable, detection of the test tube, dispensing of liquid, and dispensing of powder. To rotate the turntable a gearbox and a DC motor will be used. The purpose of a gearbox is to increase or decrease speed, where the turning force (torque) will be the inverse of the function of speed. In this case, a DC motor turns extremely quickly (12530RPM) but does not have much turning power (4.6mNm), so the gearbox will be employed to increase turning power in exchange for a slower speed. The final, slow-turning gear in the gearbox will be connected to the turntable resulting in its rotation. The detection of the test tube will be responsible for sending a signal to the motor to stop or continue rotating, for this an infrared sensor will be used. The sensor can detect if there is an object in front of it, so if there is no object the turntable continues and stops its rotation if there is one.

The dispensing of liquid will be controlled by a water pump connected to a reservoir. The pump will be connected to the dispensing head, which will have an open hole above the desired test tube, since it pumps water at a consistent rate the pump will be turned off by an Arduino code when the desired amount has been dispensed. The powder will be dispensed from a powder reservoir in the dispensing head, there will be a small hole that allows the flow of powder from the head to the test tube which will be at a constant calculated rate. The Arduino will then send a signal to a servo which can rotate about an axis that will act as a cover to the hole and will rotate in a way that covers the hole when the specified powder has been dispensed.

All the sub-processes together will result in a turntable that can dispense quantities of powder and fluid specified by the user in accurate measurements. This report will describe the technicalities behind these sub-processes that were summarized above, then will elaborate on how the sub-processes will come together to create a final product. It will explain how the model team 141 has created will be tested and modified to ensure feasibility for stakeholders and how iterations of the product could aid in creating an automated fluid dispenser that approaches perfection. [DS1] Regarding testing for electrical parts a test code will be used on them to see if they programmed and functioned as intended. Following this the turntable will be tested with the gearbox and if that functions then the powder dispensing, and water dispensing will all be tested together. The design was made to have consistent behavior to allow ease of use to stakeholders and has easy to access switches to start and stop the machine. Ultimately the testing met most of the requirements of the device, met all safety standards, as it was able to stop whenever intended and had a covered electrical compartment. Unfortunately, the prototype failed to dispense powder, and its functionality should be improved for future iterations.

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2. Problem Definition

2.1 Understanding the Problem

Opioid-related overdose has unfortunately become a leading societal concern. Since the Public Health Agency of Canada (PHAC) began surveilling opioid-related deaths in 2016, this issue has exacerbated year by year. According to the PHAC, from January 2016 to March 2023, there have been “a total of 38,514 apparent opioid toxicity deaths.” [1] In the months of January 2023 to March 2023, this has translated to 21 deaths per day. [1] With opioid-related deaths rising each year, measures must be taken to develop technologies and medicines that can combat this crisis.

2.2 Overall Product Goal & Main Function

The overall goal of this project is to design, test, and iterate an automatic fluid dispenser (AFD) for use in the pharmaceutical industry in preparing overdose countermeasures. With overdoses becoming a major cause of death around the world, the need for active pharmaceutical ingredients (APIs) that can block or potentially reverse the consequences of opioid drugs is on the rise. An example of a crucial API is naloxone hydrochloride dihydrate. Since naloxone hydrochloride dihydrate is in powder form, to deliver this API to those in need in the form of a liquid dosage, it must be dissolved in a saline solution. Considering this, the primary function of this AFD prototype will be to autonomously prepare five API solutions of arbitrary concentrations in test tubes mounted on a rotating turntable apparatus. To perform this task, the AFD must dispense a specific mass of the API in a specific volume of water.

2.3 Prototype Subfunctions

Since this entire process must be autonomous, there are many subfunctions that must be performed to achieve the overall goal. The AFD needs to analyze the input data and select the concentration of the API solution to be made. Once doing so, the turntable needs to rotate, with the first test tube in the correct position for dispensing. Next, a signal must be sent to the dispenser and a specific mass of the powder (API), and volume of water will be released. Another crucial subfunction is the AFD determining when to stop dispensing the powder (API) and water, which would be followed by a rotation of the turntable to produce the next solution. Although these are the main subfunctions of this prototype, there are many other parts and functions that come together to help achieve the end goal. See the complete function structure with energy and material conversions in Figure 11.

2.4 Design Constraints

For this design project, there is a limit on the size of the AFD, with maximum dimensions of 200 mm x 200 mm x 210 mm. With numerous groups working to develop the same prototype, there is a large demand for materials and their associated machinery. For instance, the maximum volume of 3D-printed filament that can be used is 30,000 mm³ and the maximum print time is 120 minutes. The maximum area of 6 mm thick acrylic that can be used is 600 cm². Aside from dimension and material requirements, there are multiple safety constraints in place, such as the need for a water-resistant enclosure around the electronics and a secure enclosure around the gears. Another significant constraint for this project is time, as all five test tubes of varying concentration must be filled within 120 seconds. Lastly, there are many design implementation constraints in place, such as limiting sharp corners and using the electronic parts that were provided. Taken together, these constraints have influenced all aspects of designing and iterating the AFD.

2.5 Customer Needs & Relevant Stakeholders

Knowing the precise needs of customers will help better understand how to design an effective product. This device must be able to dissolve naloxone hydrochloride dihydrate in water without the need for human interaction, be simple to construct and transport, produce varying concentrations of API solution, and be scalable. Pharmaceutical technicians also require that there are minimal wires and tubes that can be caught on, that calibration is simple, and that filling and emptying the reservoirs is easy.

Opioid overdose patients, primarily between the ages of 20 and 59 [1], are at the forefront of relevant stakeholders as the dosages of APIs produced by the AFD directly influence their survivability. As the primary customer of the AFD, pharmaceutical companies are interested in how effective, efficient, and viable it is. These companies will also be heavily invested in the practicality and scalability of the AFD to meet their immense production quotas. Health Canada and the United States Food and Drug Administration are prime examples of federal agencies that are interested in the AFD, as they have a responsibility to ensure that the dosages produced are safe for the public. Hence, these federal agencies would be interested in clinical trials to determine the efficiency and accuracy of the AFD before it can be cleared for use. For instance, Health Canada would evaluate the product to ensure that it meets the standards of section C.02.005 of Canada's Current Good Manufacturing Process, which poses requirements for the prevention of API contamination [2]. The Canadian Government would be an overarching stakeholder as the AFD helps address UN Sustainable Development Goal 3, which is to promote good health and well-being [3]. Target 3.5 of this goal is to "strengthen the prevention and treatment of substance abuse," [4] which the AFD helps address by delivering API solutions for opioid overdose patients. Lastly, the AFD will directly increase health-adjusted life expectancy [5] and help meet unmet healthcare needs [5], thus addressing the health section of the government's Quality of Life Framework for Canada. [5]

2.6 Safety Risks & Mitigation

It is crucial to consider the safety risks associated with this project design. As the AFD is dispensing water into the test tubes, there is a risk of electrocution caused by water encountering the electronic components. To prevent such risk, the Arduino Uno, motor shield, breadboard, and electric motor will be housed in a sealed, water-resistant acrylic container. Other electronic components that need to be located elsewhere, such as the IR sensor or servo, will be kept clear of the water reservoir. All wiring to the dispensing head will travel through the inside of the mast directly from the electronic storage container to isolate the wiring from any water source. Another safety hazard is exposed tubing and piping getting caught on unwanted objects, which could result in the AFD tipping over and falling and in the dispersal of the API onto workers. To mitigate this risk, all external tubing from the water reservoir will be clamped tightly to the surface of the AFD. All wires connecting the Arduino to the servo in the mast head will be enclosed within the mast wall, preventing any interaction with the outside environment. If an unforeseen short-circuit or error were to occur, to prevent this error from imposing risk to the user, the AFD will have an emergency stop button that cuts power to the device.

3. Prototype & Testing

3.1 Key Decisions Leading to the Prototype Design

It was determined by the team that the height of the AFD would be the most important dimension to consider as the length and width could be easily accommodated. A unique design process was used, where unchangeable and restrictive component heights were first identified to understand how much of the maximum height was remaining for the more flexible components.

For example, the test tubes provided were approximately 107 mm tall, so this meant that the dispensing head, gearbox, and the clearance between the dispensing head and test tube had to fit within the 103 mm that remained of the maximum 210 mm height. Hence, the team determined that the AFD would be 210 mm tall since the height needed to be maximized. The next step was to design the geartrain because its vertical height would influence the gearbox height and the height of the dispensing head. The gear train was a highly restrictive component as it was not as easily changeable due to torque and speed requirements. Using the mathematical modelling, a gear train with a gear ratio of 250 was designed, and the gearbox height was determined to be 48.8 mm. Hence, 54.2 mm was left for the dispenser head and the clearance between it and the test tube. The dispenser head was designed to have a height of 40 mm to allow enough space for the servo to be housed and for a built-in powder reservoir. Thus, leaving an 8.8 mm gap between the bottom of the dispenser head and the test tube. Once the vertical dimensioning was complete, the rest of the components were horizontally mapped out on a 200 mm by 200 mm plane.

The IR sensor was mounted inside the mast head so that it was situated as close as possible to the top of the test tube for detection. The servo motor was mounted inside the mast head since it was used to control the dispensing of the API from the built-in 3D-printed powder reservoir. The electronics control box was situated at the base of the mast so that wiring could be run up the inside of the acrylic mast up to the IR sensor and servo motor housed in the mast head. This allowed for convenient access to the wiring while providing a waterproof housing. Additionally, by situating the electronic control box beside the gear box, this allowed for hole to be cut in the shared wall between the gearbox and electronic control box for the motor to be mounted and wired. Not to mention, this protected the motor from any water spillage. A plastic water bottle was used as the water reservoir to save materials and provide waterproof containment.

3.2 The Prototype

The prototype that team 141 built consists of multiple detachable parts, these parts each come together to create the final prototype and each one represents their own sub processes and purpose. The main 3 components that make up the final prototype were the gearbox, the mast, and the masthead. The gearbox consists of a wooden box that which houses the gears necessary for the turning function of the turntable. The mast has acrylic and wooden supports which connect to create a strong foundation, the inside of the mast was hollow so that tubing and wires could flow freely throughout. The mast head's primary purpose was to have holes designed for tubing for water dispensing, the infrared sensor to sense rotating test tubes and for a funnel which would dispense powder. The left and right side views of the prototype can be seen in Figure 1 and Figure 2.

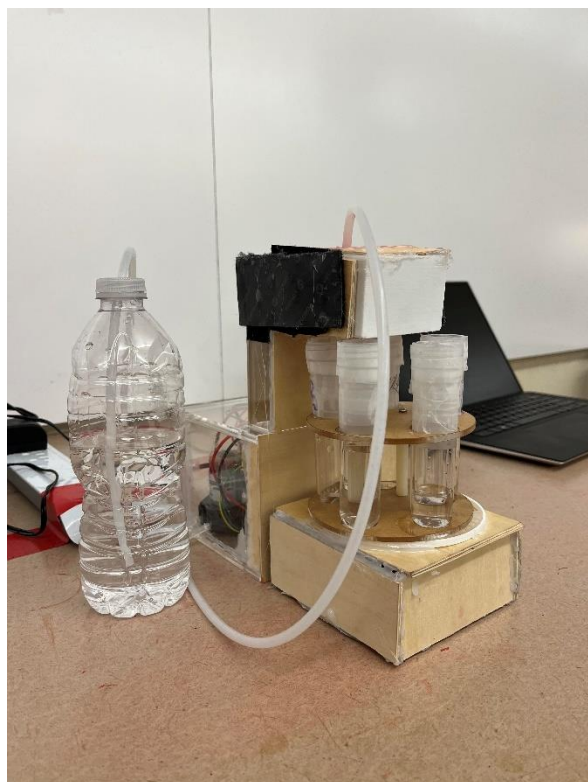


Figure 1: Left auxiliary view of complete prototype.

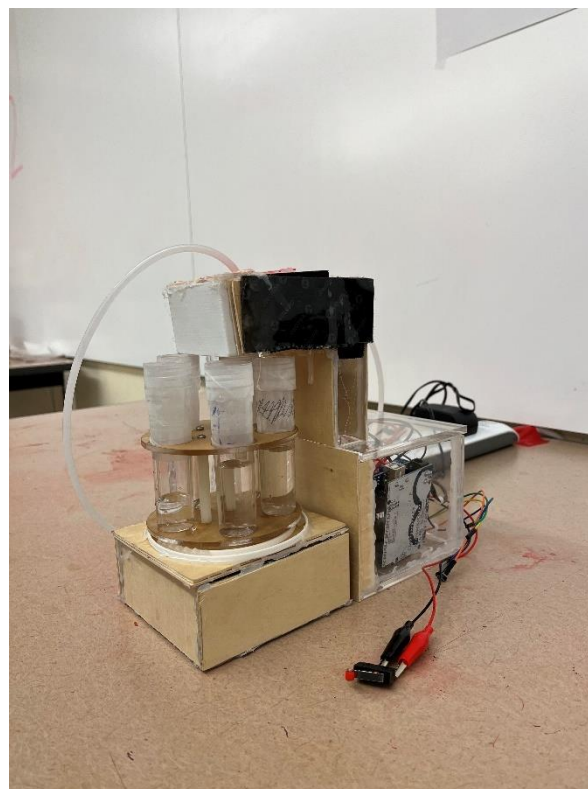


Figure 2: Right auxiliary view of completed prototype.

3.2.1 The Gearbox

The gearbox consists of a 100x100x48.8mm wooden box which contains the gears required to increase the mechanical advantage of the motor. One of the walls of the box was made of acrylic with a friction fit hole for the motor. The motor was fit into a specific orientation to ensure that it is properly aligned with the bottom gear permitting rotation and was glued to prevent its movement. On the inside of the gearbox there is an elevated acrylic support so that there would be a shaft that will peek through the top of the gearbox allowing for the rotation of the bottom rack. The lid of the gearbox had a hole in it so that the 20-tooth gear at the top of the elevated shaft could go through the lid and be glued to the bottom rack so it would spin along with the gears connected to it. The top of the lid also has a track that runs along the outsider perimeter of the lid with the gap between inner and outer rings being 6.2 mm to house ball bearings. Ultimately the bottom rack would rest both on the 20-tooth gear at the top of the gearbox as well as the ball bearings at the same time, the primary purpose of this was to reduce friction and to disperse the load so that all the force would not be centered on the 20-tooth gear. To ensure the bottom gear shaft does not spin out of its hole from the fast rotation of the worm gear attached to the motor an L shaped support was installed into the inside of the gearbox preventing the bottom shaft from spinning

out. The L-shaped support with the inside of the gearbox can be seen in Figure 4 and the overall gearbox part including the racks and ball bearing track can be viewed in Figure 33.2.2 The Mast.



Figure 3: Overview of Gearbox with connected ball bearing track, and bottom rack.



Figure 4: Inside view of gearbox showing alignment of motor with gears and L-Shaped support.

The mast consists of 2 main acrylic and wooden supports which were used to enclose electronic wiring throughout and fortify the general structure. The mast is hollow and at the base is the electronic box which will have an opening so that the dc motor from the gearbox will fit in allowing the electronics to be covered and connected. The top of the mast has the hook side of Velcro attached to the acrylic pieces which will allow a Velcro connection with the mast head. There are also small wooden supports that come out from the top of the mast to stop the mast head from tilting forward as its weight and extension would otherwise not have enough support causing a forward tilt. As seen in Figure 5 the hook portion of the Velcro were attached to the acrylic sides of the mast, and there are supports coming out of the mouth of the structure to eventually support the mast head.



Figure 5: Mast with wooden supports coming out of the mouth of the structure, to stabilize the mast head.

3.2.2 The Mast Head

The mast head is a small rectangular prism piece that had three holes for various functions, one hole had a funnel made of paper which fit into it to allow for the dispensing of powder at a specific rate and to ensure the powder dispenses at a rate slow enough to calculate and predict. The pump would siphon water from a water bottle with tube fit hole through the bottle cap to fit the tubing allowing the bottle to be used as a water reservoir for the system. After the pump siphons water from the reservoir the tubing comes up through the hollow part of the mast and would be fit into its respective hole in the mast head allowing the water pump to dispense the water to the test tubes. The final hole was used for an IR sensor to detect whether a test tube was under the dispensing head. The sides of the mast head have the hook side of Velcro, so that when the mast head and mast are aligned one piece of loop on either side Velcro could connect the two as seen in Figure 6. The two holes for the water and powder can be seen in Figure 6 however the IR sensor is hidden underneath these two as it needs to be extremely close to the tube to detect them.

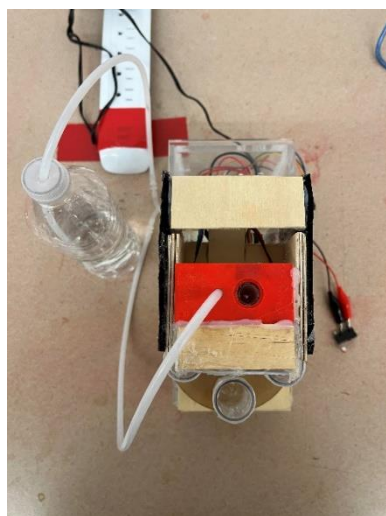


Figure 6: Image of mast head holes and water reservoir with tubing.

3.3 Prototype Testing

There were 4 tests that were conducted to test the ability of the prototype, that is, to test the feasibility/safety and the overall function of the automated fluid dispenser. The first test was a safety test to see if the prototype would be able to stop whenever it was intended. During the testing, the prototype was not able to complete this function. However, the emergency stop was working prior to the testing when the team conducted their own tests for the video submission, and so there may have been wires that had disconnected on the way to testing which could have caused it to malfunction. The next test that was conducted on the prototype was to test if the prototype could dispense a specified amount of water within 1-2 mL (1mL for full marks). The prototype created by team 141 was able to dispense the specified amount of water to all the test tubes within an error of $\pm 2\text{mL}$ as it was expected to dispense 15mL and only dispensed 13mL. This was extremely reliable as the rotation of the test tubes functioned as expected, perfectly lining up at the right time for the water dispensing. The reason that it may not have dispensed 15mL exactly may be a result of systematic uncertainty, as the team was not provided the tools to accurately measure 15 mL beforehand and ultimately was based upon less accurate measurements. Despite the dispensing not being accurate it would consistently be inaccurate by the same amount, this

consistency in inaccuracy means that it could be fixed by slightly changing the time that the pump would be active for in the code. The next test would be to dispense a certain volume of powder into the test tubes, the prototype that was made did not pass this test. The prototype was unable to dispense the powder consistently or reliably, as the IR sensor could not reliably detect test tubes for this function. The next test was a deflection test, to see if the mast head could withstand a lateral force of 5N without deflecting more than 10 mm, the prototype passed this test very easily as there was almost no movement in the masthead. As a result the prototype would reliably and consistently pass this test.

3.4 Evaluating Mathematical Models

3.4.1 The Gearbox

Since the infrared sensor was inconsistent when detecting the tubes, the rotation of the turntable had to be hard coded, this meant that the amount of time for a rotation of the turntable must be calculated to find how long the gearbox should run until the next test tube is under the dispensing head. After marking one of the test tubes, the gearbox ran for several rotations and was recorded to find the amount of time it takes for each rotation of the turntable. After averaging the turns, the time per rotation came out to be 1.3 seconds per rotation which when converted to rotations per minute is 46.15 RPM. The RPM predicted by the mathematical model after expected losses to friction was 40.5 RPM as seen in sample calculation using equation (1) below.

Table 1: List of variables, descriptions necessary for finding errors in past mathematical models.

Variables	Description
ω_{out}	Angular velocity of final gear in gear train (effectively angular velocity of turntable)
$\omega_{post-Fric}$	Angular velocity of final gear in gear train (effectively angular velocity of turntable) after expected friction losses
μ	Friction multiplier on ω_{out} which depends on expected percent loss to friction and number of gear stages (a 10% loss to friction per stage would result in $\mu = 0.9$)
S	Number of gear stages

$$\omega_{post-Fric} = \omega_{out}(\mu)^S \quad (1)$$

$$\omega_{Post-Fric} = 50RPM(0.9)^2 = 40.5 \text{ RPM}$$

The calculated result was an underestimation of the true turning speed of the turntable; the expected loss was 10% per gear train which would end up being a multiplier of 0.9^2 on the base turning speed. This means that the expected loss should be lower, and through calculation taking the turning speed before losses as 50 RPM the expected loss due to friction per gear train is approximately 4%. Errors in torque calculations cannot be evaluated from the turning of the turntable itself as it successfully turned meaning

the calculated torque was at least greater than the amount required. This can contribute to uncertainty as it does not allow for fine tuning to see how much less torque can be used, however in the case of an automated fluid dispenser speed is not a desired function meaning that after a certain speed threshold which could arbitrarily be anything above 15 RPM, reducing torque has no function for the purpose of an AFD.

$$50x^2 = 46.15$$

$$x = 0.96$$

3.4.2 Mast Deflection

According to the mathematical models described and used in the team's 9T report, the maximum lateral deflection of the mast was calculated to be 0.191 mm and 4.19 mm for the two possible directions from which the force could have been laterally applied. As described in the Prototype Testing section above, the team's prototype successfully passed the deflection test with the mast not experiencing any deflection at all when a lateral force of 5N was applied. When evaluating the deflection approximations, it is evident that they were overestimates of the actual deflection. This can be attributed to the fact that when the deflection was calculated, the only supports used in the calculation were the two acrylic beams. However, as seen in Figure 2, the team added two 1.5 mm thick wooden panels to the mast structure to create an enclosed housing for the wires. Therefore, these two wooden panels would have added further structural rigidity that was not accounted for in the deflection calculations. Not to mention, the friction between the acrylic beams and the lid of the electronics control box that they passed through, and the lack of mobility of these beams due to this fixed set-up were not accounted for. This explains why the team's deflection calculations were an overestimate and why when tested, the mast structure did not deflect at all despite the calculations suggesting otherwise.

3.5 Evaluating the Prototype

3.5.1 Prototype Functionality

As outlined in the Overall Product Goal & Main Function section of the Problem Definition, the primary functional purpose of the prototype is to autonomously prepare five API solutions of arbitrary concentrations in test tubes mounted on a rotating turntable apparatus. To perform this task, the AFD must dispense a specific mass of the API in a specific volume of water. As described in the Prototype Testing section above, the team's AFD prototype was able to dispense water into the five test tubes with an error of ± 2 mL in contrast to the amount that was specified. With the turntable stopping perfectly below the mast head in each revolution and the water being dispensed into each tube with no spillage, it can be concluded that the functionality for the dispensing of the liquid was successfully achieved by the team. Conversely, the second key functionality of the AFD, the dispensing of the API, was not achieved by the team's AFD prototype. The prototype failed this test as it was unable to dispense the powder reliably, accurately, or consistently into each test tube. Thus, it can be evaluated that the prototype only successfully fulfilled half of its functionality targets as it was able to accurately dispense a specific volume of water into the five rotating test tubes, however failed in the dispensing of the API solution. The 4.1.4 Optimized IR Sensor Positioning section addresses the changes the team recommends moving forward to ensure that this functionality is successfully achieved since dispensing an accurate amount of the API into the five tubes is integral to preparing overdose countermeasures for opioid overdose patients.

3.5.2 Evaluation of the Prototype Using Design Constraints

Outlined in the Design Constraints section of the Problem Definition are some of the key design constraints for the AFD prototype, the majority of which, team 141's prototype met. To begin, the final width and height dimensions of the prototype were within the constraints, with a width of 180 mm and a height of 210 mm. However, when the prototype was assembled, the length of the prototype was 230 mm, which is 30 mm over the length limit of 200 mm. This can be attributed to the use of two wood panels for the walls of the gearbox and hot glue between joints, which added to the designed length of 200 mm. Nonetheless, with a length of 230 mm, the prototype failed to meet this design constraint. As for the material constraints, the team's prototype met all requirements, with a total 3D print volume of 28,980 mm³ and 599 cm² of 6 mm thick acrylic used. All electronics, such as the Arduino, water pump, motor, breadboard, and motorshield, were successfully stored in a water-resistant enclosure with the electronics control box being completely enclosed using an acrylic structure and hot glue closing all seams and openings. Additionally, wiring to the mast head from the electronics box was guided through the inside of the mast, which was also made of acrylic and completely waterproof. Conversely, the wiring for the start button and emergency stop button were not enclosed in a waterproof housing because as seen in Figure 2, the wiring for these buttons was outside of the enclosure. The team has addressed this issue in the 4.1.6 Start and Emergency Stop Button Wiring & Water Tubing section below. The servo motor and IR sensor were mounted in the 3D-printed dispenser head, which was sealed and not prone to any water entering its inside housing. Although the prototype failed in dispensing the API powder, it was able to successfully dispense the water in the five rotating test tubes within the 120 second limit. Lastly, all electronic parts used were those provided.

3.5.3 Evaluation of the Prototype Using Customer Needs

The Customer Needs & Relevant Stakeholders section of the Problem Definition describes the key needs of the customer's of the AFD prototype. To begin, the prototype must be simple to construct and transport. The team's prototype fulfills this need via the design to separable subsystems. For example, the geartrain, mounted turntable, and test tube rack are easily slidable from underneath the mast head and are fixed to the gearbox structure. This allows for these three components to be easily removable as they can be transported as one. The use of Velcro to attach the mast head to the mast structure allows for simple and easy installation and deconstruction, with all electronic wires being easily removable and stored in the mast structure itself. The mast structure is fixed to the electronics control box and therefore, does not need to be assembled. The electronics within the control box can be conveniently accessed via the sliding acrylic top. Lastly, the water reservoir can simply be picked up and the tubing can quickly and easily be attached. Filling and emptying the water reservoir is convenient as the cap can be unscrewed and water can be added via a tap and removed by inverting the reservoir. Likewise, the powder reservoir can be easily filled via the cap on the top and removed by detaching the mast head and inverting it. One key customer need that the prototype fails to perfectly meet is having minimal wires and tubes that can be caught on. Although all wiring is held within the electronics control box and run through the inside of the mast to the mast head, as seen in Figure 2, the wiring for the start and emergency stop buttons is outside of the housing and can get caught on and damage the electronics. Additionally, the water tubing from the top of the water reservoir to the mast head is not taped down and it is possible that this tubing could get caught on and cause damage to the prototype. Both possible safety risks will be addressed in the 4.1.6 Start and Emergency Stop Button Wiring & Water Tubing section.

4. Final Design

4.1 Design Improvements & Modifications for Optimized Turntable, Gearbox & Dispensing Systems

Through the process of assembling the turntable and dispensing systems, the team noticed many facets of the design which could be improved upon. The testing revealed that the sizes of the electronic box and gearbox could be better optimized, the water reservoir could be altered, and that the powder dispensing could be made more accurate.

4.1.1 Material Iterations

In the next iteration, the materials of both the gearbox and water reservoir should be changed to 3D printed PLA. This change will not only ensure that the gearbox is housed in a waterproof enclosure but will also allow for precise dimensioning. The team noticed that when the gearbox was constructed using the wooden panels, it was difficult to cut the panels to the exact dimensions that were needed as precise cutting machinery was unavailable. Consequently, the gearbox was not level and this interfered with the movement of the ball bearings in the track. Additionally, some walls had to be made of two wooden panels due to the lack of the materials strength which contributed to the prototype not meeting the length constraint as described in the Evaluation of the Prototype Using Design Constraints section. Designing the gearbox on SOLIDWORKS will ensure accurate dimensioning and that the prototype meets the size constraints. 3D printing the water reservoir would also be advantageous as it can be dimensioned to hold the necessary volume of water and reduce the total space that the prototype occupies. In the team's prototype, a 500 mL plastic water bottle was used as the water reservoir. However, as per the design constraints checklist, the prototype only needs to hold a maximum of 100 mL of water. Thus 3D printing the water reservoir will allow for a smaller, more efficient reservoir to be designed and constructed that will reduce the space that the prototype occupies. This will also allow for the water reservoir to be mounted to the mast, allowing for the tubes to be taped down and addressing the safety risks mentioned in the Evaluation of the Prototype Using Customer Needs section.

4.1.2 Supports for the Geartrain and Gearbox Height

Another alteration is redesigning the geartrain to reduce its vertical height while maintaining the gear ratio necessary to perform the required functions. This will lower the height of the gearbox, therefore, reducing the mast structure itself and saving materials.

Inside the gearbox, adding supports to the lower gear shaft is imperative because during the team's independent testing of the prototype, the shaft hole that was designed became loose over the course of numerous tests. Subsequently, the shaft would slip out of its place, requiring adjustments. The team had to improvise iterations to solve this issue, which can be seen with the use of the wooden L bracket in Figure 4 in the 3.2.1 The Gearbox section above. That being said, Figure 9 below offers a new iteration of added support at the bottom of the shaft.

4.1.3 Dimensioning of the Electronics Control Box

Another important iteration is increasing the width of the electronics box while reducing its length, which will make better use of the size of the peg board underneath. As mentioned in the Evaluation of the Prototype Using Design Constraints section, the team's prototype was 30 mm over the length constraint. By reducing the length of the electronics box and increasing its width, the extra space in the width can be

optimized while ensuring that the length constraint is fulfilled. This optimization of space is illustrated in Figure 7 below.

4.1.4 Optimized IR Sensor Positioning

The location of the infrared sensor is a key iteration, with it now moved to a vertical orientation and mounted to the mast structure as seen in Figure 10 below. This new positioning of the IR sensor will ensure that it is closer to the test tubes, making detection easier and more reliable compared to when it was mounted in the dispensing head. The team believes that this new orientation will ensure that the test tube locations are detected more efficiently and accurately, and thus, the powder dispensing mechanism would be more accurate and reliable as well.

4.1.5 Optimized Mast Head Attachment

In the team's prototype, the dispensing head was connected to the mast structure via Velcro. Although this mechanism allowed for the easy removal of the mast head and access to the wiring inside the mast, it was not as aesthetically pleasing for a professional prototype. A better alternative that maintains the same functionality as the Velcro but offers a professional and clean appearance is screwing the mast head to the mast structure, as illustrated in Figure 8 below. Doing so will provide a more secure and seamless fit while being cost-effective. Finally, to secure the components on the peg board, the team added pegs to all components resting on the peg board, which can be seen in Figure 9 below.

4.1.6 Start and Emergency Stop Button Wiring & Water Tubing

As outlined in the Evaluation of the Prototype Using Design Constraints and Evaluation of the Prototype Using Customer Needs sections above, the wiring for the start and emergency stop buttons was outside of the waterproof electronics control box. Since this poses a safety concern for both the operators of the prototype and the electronics, these wires will be housed inside the electronics control box and the two buttons will be mounted in the wall of the box instead of resting on the peg board. Additionally, these two sections acknowledged that the water tubing was loose and could get caught on, and therefore, all water tubes will be taped down to the side of the mast structure.

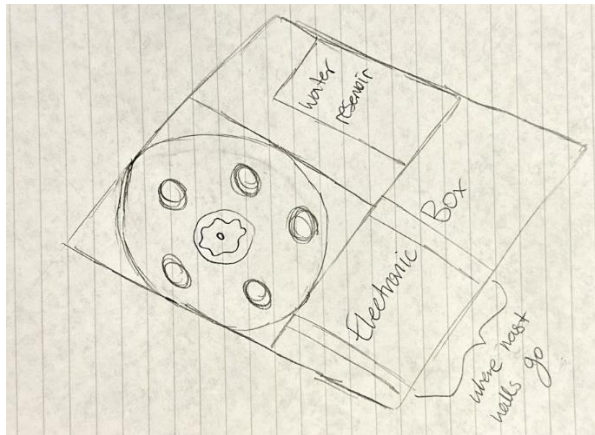


Figure 7: Top-down view of new design layout. The electronic box takes up a larger part of the area with an increased width and reduced length. The water reservoir has been moved accordingly.

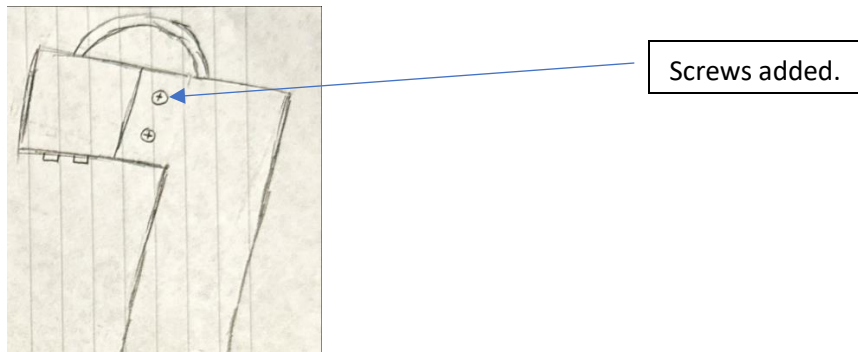


Figure 8: Side-view of mast head and mast structure interface, now attached via screws instead of Velcro.

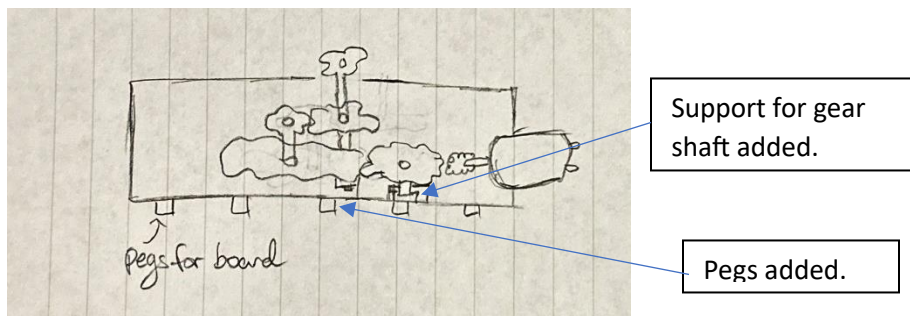


Figure 9: Side-view of new gearbox design, now including pegs on the bottom and new support for the bottommost gear shaft.

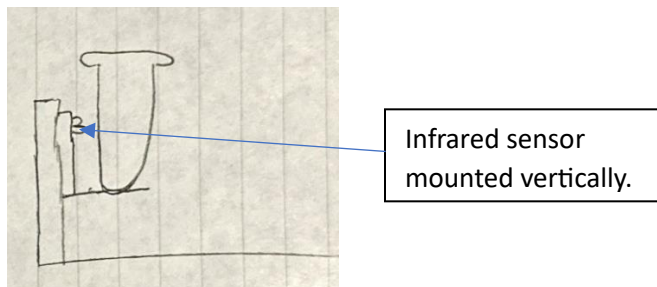


Figure 10: Side-view of the new IR sensor position, mounted vertically on the mast structure next to the turntable to better detect tubes passing by.

4.2 Considerations for Safety, Sustainability, & User Accessibility in Material Design & Implementation

Many factors were considered when designing and implementing ideas related to materials and the safety of both the components and the operators of the product. For instance, the electronic box is encased in water-proof acrylic making interaction with water unlikely and dangers associated with the handling of the machine reduced. In terms of sustainability, the parts used for the encasing of the gearbox, electronics box, water reservoir, and mast head will be made from either acrylic or PLA making reusability possible and the need for replacement of parts unnecessary. To make sure the device is accessible and user-friendly, the team focused on making the operation of the machine simple to use by beginning the operation of the dispensing by the press of a single button and the disassembly process easy.

Team 141's design incorporates several safety features, ensuring the protection of the user. An emergency stop button is included to stop all operations in case of an emergency or desired stop. The electronic

components are securely housed in a waterproof acrylic box to prevent any electrical hazards and both the acrylic and PLA used in the design is non-toxic and safe. The design minimizes the risk of cross-contamination, a crucial aspect when dealing with active pharmaceutical ingredients by making a specific reservoir for powder with a lid and pumping the water via a plastic tube. The dispensing head is easy to clean and replaceable if necessary. Adherence to these standards and regulations ensures that the device is safe, efficient, and accessible.

5. References

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- [6] T. Rich, "Yamaha Music," Yamaha Corporation of America and Yamaha Corporation, 11 August 2021. [Online]. Available: <https://hub.yamaha.com/audio/vinyl/heres-what-you-need-to-know-about-vinyl/#:~:text=Vinyl%20records%20come%20in%20three,to%20spin%20on%20the%20turntable..> [Accessed 5 November 2023].

6. APPENDICES

6.1 APPENDIX I: Work distribution

Task	Description of Activity	Activity Duration (hours)	Individual Responsibility for Activity
3.2 The Prototype	Writing about how the parts of the prototype fit together and collecting images of prototype and its constituent parts.	1.5	Nolan
3.3 Prototype Testing	Writing about the results of the prototype testing, the consistent, inconsistencies and things to work on in the future for a complete prototype	1.5	Nolan
3.4.1 The Gearbox	Writing about any errors in calculation of the mathematical model, assumptions that were incorrect or need changed.	1	Nolan
1.Executive Summary	A summary of the report that a non-technical reader can grasp.	1	Nolan
4. Final Design	A description of the iterations and alterations made to previous versions of the device to come up with a final design.	2.5	Rafael
2.Problem Definition	Discussed all sections, including: The Real Problem, Overall Product Goals & Main Function, Subfunctions, Design Constraints, Customer Needs & Relevant Stakeholders, and Safety Risks & Mitigation.	0.5	Ishaan

3.1 Key Decisions Leading to the Prototype Design	Provided a summary of the key considerations and important decisions the team made that resulted in the final prototype design.	0.5	Ishaan
3.4.2 Mast Deflection	Evaluating the team's deflection calculation approximations regarding the actual deflection. Identified assumptions and possible errors in calculation that led to discrepancies.	0.5	Ishaan
3.5 Evaluating the Prototype	Including all subsections. Evaluated the team's prototype with respect to the functional requirements, design constraints, and customer needs.	2	Ishaan
4.1. Design Improvements & Modifications for Optimized Turntable, Gearbox & Dispensing Systems	Descriptions of the iterations the team would make.	1.5	Ishaan
Editing	Editing the report	1	Ishaan
6.2 Appendices	Things learnt during this design project and things that will be done better in next project	1	Amarachukwu

6.2 APPENDIX II: Team Performance Assessment

The process of creating this design helped the team recognise and acquire abilities and obstacles that affect the way decisions are made when tackling problems. They include the unconscious bias that skews rational problem-solving, the knowledge of how to design solutions that satisfy stakeholders and meet constraints, the ability to provide constructive criticism and feedback, the right way to share one's ideas with the team, Canadian engineering ethics and principles, and the significance of time management and planning.

The team will work to ensure that the next project is better by providing a more explained testing plan and using language appropriate for the target audience. In case the team's next project involves designing

an automatic fluid dispenser, a low gearbox height will be used to allow the IR sensor to be positioned outside the dispensing head and to allow more space between the dispenser's head and the test tube, 3D printing will be used to make the gearbox and water reservoir aesthetically pleasing and easy to use, screws to secure the dispensing head to the shaft to ensure minimal head deflection, a funnel-shaped structure to hold and dispense the powder, and extend the design's backside (which stores the electronics) to ensure the design stays within the specified bounds.

6.3 APPENDIX III: Arduino code

```
#include <AFMotor.h>
#define IRpin A0
#define pushButtonForwards A1
#define limitSwitch A3

AF_DCMotor DCMotor(1);
AF_DCMotor pump(4);

bool pushButtonStateForwards = HIGH;
bool pushButtonStateBackwards = HIGH;
bool limitSwitchState = HIGH;

int counter = 0;

void setup() {
  Serial.begin(9600);

  DCMotor.setSpeed(200);
  DCMotor.run(RELEASE);

  pinMode(pushButtonForwards, INPUT_PULLUP);
  pinMode(limitSwitch, INPUT_PULLUP);
  pinMode(IRpin, INPUT);
}

void loop() {
  pushButtonStateForwards=digitalRead(pushButtonForwards);
  limitSwitchState=digitalRead(limitSwitch);

  if (pushButtonStateForwards==LOW && limitSwitchState==HIGH){

    // first run
    pump.run(FORWARD);          // Change between FORWARD or BACKWARD (all
upper case, Case sensitive) depending on what directon the liquid flows in the
pump.
    pump.setSpeed(255);          // Set the speed of pump. Uses an 8 bit
value, 255 = 100% speed.
```

```

    delay(15000);                // Choose how long the pump should run,
    depending on how much volume to fill.
    pump.run(RELEASE); // Stop the pump
    delay(1000);
    Serial.println("forwards");
    DCMotor.run(FORWARD);
    delay(250);
    DCMotor.run(RELEASE);

    // second run
    pump.run(FORWARD);           // Change between FORWARD or BACKWARD (all
    upper case, Case sensitive) depending on what directon the liquid flows in the
    pump.
    pump.setSpeed(255);          // Set the speed of pump. Uses an 8 bit
    value, 255 = 100% speed.
    delay(15000);                // Choose how long the pump should run,
    depending on how much volume to fill.
    pump.run(RELEASE); // Stop the pump
    delay(1000);
    Serial.println("forwards");
    DCMotor.run(FORWARD);
    delay(250);
    DCMotor.run(RELEASE);

    // third run
    pump.run(FORWARD);           // Change between FORWARD or BACKWARD (all
    upper case, Case sensitive) depending on what directon the liquid flows in the
    pump.
    pump.setSpeed(255);          // Set the speed of pump. Uses an 8 bit
    value, 255 = 100% speed.
    delay(15000);                // Choose how long the pump should run,
    depending on how much volume to fill.
    pump.run(RELEASE); // Stop the pump
    delay(1000);
    Serial.println("forwards");
    DCMotor.run(FORWARD);
    delay(250);
    DCMotor.run(RELEASE);

    // fourth run
    pump.run(FORWARD);           // Change between FORWARD or BACKWARD (all
    upper case, Case sensitive) depending on what directon the liquid flows in the
    pump.
    pump.setSpeed(255);          // Set the speed of pump. Uses an 8 bit
    value, 255 = 100% speed.

```

```

    delay(15000);                // Choose how long the pump should run,
    depending on how much volume to fill.
    pump.run(RELEASE); // Stop the pump
    delay(1000);
    Serial.println("forwards");
    DCMotor.run(FORWARD);
    delay(250);
    DCMotor.run(RELEASE);

    // fifth run
    pump.run(FORWARD);           // Change between FORWARD or BACKWARD (all
    upper case, Case sensitive) depending on what directon the liquid flows in the
    pump.
    pump.setSpeed(255);          // Set the speed of pump. Uses an 8 bit
    value, 255 = 100% speed.
    delay(15000);                // Choose how long the pump should run,
    depending on how much volume to fill.
    pump.run(RELEASE); // Stop the pump
    delay(1000);
    Serial.println("forwards");
    DCMotor.run(FORWARD);
    delay(250);
    DCMotor.run(RELEASE);
}
else if (limitSwitchState==LOW){
    DCMotor.run(RELEASE);
    pump.run(RELEASE); // Stop the pump
}
delay(50);
}

```


6.4 APPENDIX IV: Function Structure Mind Map

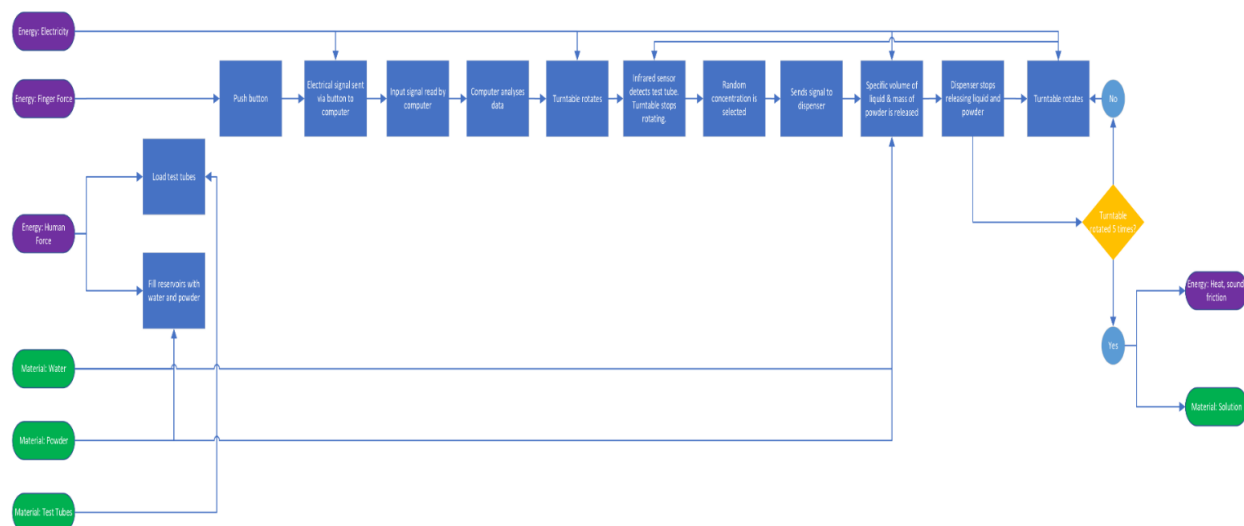


Figure 11: Function structure mind map displaying subfunctions and energy and matter conversions.